

Report Laboratories 8

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Source code

Problem Description

Given a set of nodes, each having a set of coordinates (x, y) and inherent cost, pick exactly half of the nodes to form a Hamiltonian cycle.

The goal is to minimize the sum of total path length plus the total cost of the selected nodes.

In this laboratory the goal was to investigate the global structure of the search landscape by inspecting correlation between similarity and quality of different variants of solutions.

Result Comparison (precious laboratories)

Algorithm	TSPA Cost (Mean (Min, Max))	TSPB Cost (Mean (Min, Max))
Random	265,135 (241,347, 291,966)	213,771 (190,834, 241,363)
Nearest Neighbor #1	85,108.5 (83,182, 89,433)	54,390.4 (52,319, 59,030)
Nearest Neighbor #2	73,601.7 (71,868, 75,953)	48,723.2 (44,609, 57,315)
Greedy Cycle	72,646.4 (71,488, 74,410)	51,400.6 (49,001, 57,324)
Nearest Neighbor Regret	117,516 (107,945, 128,071)	73,512.7 (67,345, 78,889)
Greedy Cycle Regret	115,137 (106,052, 123,750)	73,316.1 (67,729, 77,498)
Nearest Neighbor Regret Weighted (0.5, 0.5)	73,566.7 (70,894, 75,929)	49,847 (44,901, 57,036)
Greedy Cycle Regret Weighted (0.5, 0.5)	72,137.6 (71,108, 73,395)	50,827.2 (47,144, 55,700)
Nearest Neighbor Regret Weighted (0.3, 0.7)	75,562.1 (72,155, 78,661)	51,784.2 (47,543, 59,409)
Greedy Cycle Regret Weighted (0.3, 0.7)	73,010.6 (70,475, 75,365)	52,420.1 (49,944, 56,179)
Nearest Neighbor Regret Weighted (0.7, 0.3)	73,412.8 (71,519, 75,234)	48,816.6 (45,153, 53,425)
Greedy Cycle Regret Weighted (0.7, 0.3)	72,437.2 (71,163, 73,759)	51,173.3 (49,326, 53,437)
Local Search Greedy + Intra Nodes + Random Start	83,804.35 (77,418, 90,972)	59,076.75 (52,869, 67,067)
Local Search Greedy + Intra Nodes + Greedy Start	72,806.49 (71,034, 74,987)	45,468.38 (43,989, 51,040)
Local Search Greedy + Intra Edges + Random Start	73,387.63 (71,248, 78,480)	48,161.21 (45,907, 51,461)
Local Search Greedy + Intra Edges + Greedy Start	71,067.07 (70,046, 73,486)	45,053.77 (43,947, 50,319)
Local Search Steepest + Intra Nodes + Random Start	87,883.71 (79,593, 98,325)	62,947.23 (54,122, 71,598)
Local Search Steepest + Intra Nodes + Greedy Start	72,807.32 (71,034, 74,904)	45,414.50 (43,826, 50,876)
Local Search Steepest + Intra Edges + Random Start	73,938.93 (71,428, 77,903)	48,323.99 (45,670, 51,667)
Local Search Steepest + Intra Edges + Greedy Start	70,975.96 (69,864, 73,068)	44,974.89 (43,921, 50,319)
Candidate Local Search Steepest (k=10)	80,070.30 (75,097, 85,343)	49,548.59 (46,836, 51,921)
Candidate Local Search Steepest (k=15)	76,673.52 (72,509, 80,911)	48,723.89 (45,961, 51,644)
Candidate Local Search Steepest (k=20)	75,295.08 (72,316, 79,297)	48,457.54 (45,886, 51,560)
Steepest with LM + Intra Nodes + Random Start	87,917.41 (80,030, 99,187)	62,530.10 (55,192, 69,603)
Steepest with LM + Intra Edges + Random Start	73,486.72 (71,347, 76,128)	48,105.17 (45,528, 51,218)
Multiple start local search (Steepest, Random init)	71,330.35 (70,705, 71,787)	45,701.80 (45,701, 46,166)
Iterated local search	69,182.00 (69,095, 69,700)	43,546.80 (43,446, 43,963)
Large Neighborhood Search (without LS)	69,499.85 (69,207, 70,161)	44,167.30 (43,671, 45,013)
Large Neighborhood Search (with LS)	69,327.95 (69,207, 69,412)	43,976.80 (43,797, 44,320)

Algorithm	TSPA LS runs (Mean (Min, Max))	TSPB LS runs (Mean (Min, Max))
Iterated local search	3292.75 (2323, 4323)	3391.95 (2052, 4654)
Large Neighborhood Search (without LS)	1380.90 (1107, 1465)	1428.20 (1391, 1450)
Large Neighborhood Search (with LS)	906.30 (888, 924)	990.55 (960, 1017)

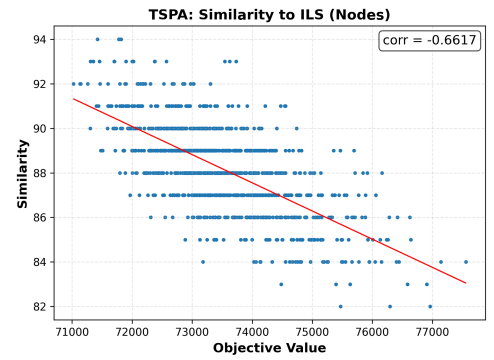
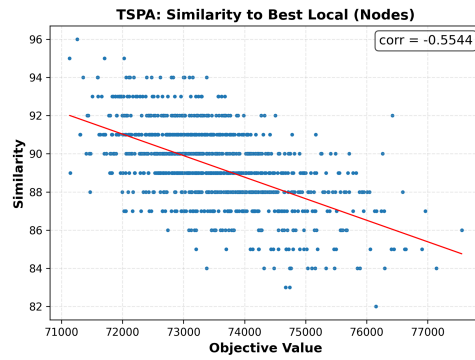
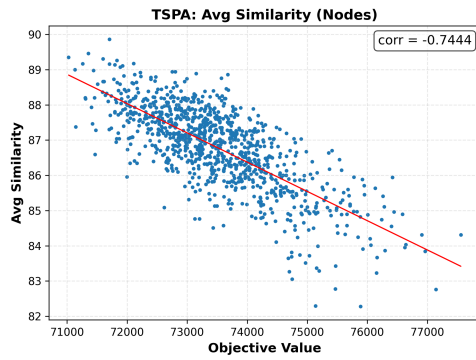
Execution Time Comparison

Algorithm	TSPA Time (Mean (Min, Max)) [ms]	TSPB Time (Mean (Min, Max)) [ms]
Local Search Greedy + Intra Nodes + Random Start	36.36 (16.00, 69.92)	35.84 (19.78, 59.54)
Local Search Greedy + Intra Nodes + Greedy Start	2.49 (1.80, 5.13)	2.67 (1.85, 7.62)
Local Search Greedy + Intra Edges + Random Start	48.81 (23.56, 70.97)	42.71 (22.79, 62.96)
Local Search Greedy + Intra Edges + Greedy Start	3.28 (1.95, 5.24)	2.92 (1.77, 7.81)
Local Search Steepest + Intra Nodes + Random Start	21.89 (16.96, 31.65)	22.20 (17.60, 29.57)
Local Search Steepest + Intra Nodes + Greedy Start	2.10 (1.55, 3.00)	2.31 (1.81, 4.95)
Local Search Steepest + Intra Edges + Random Start	15.45 (13.59, 19.09)	15.45 (13.17, 17.90)
Local Search Steepest + Intra Edges + Greedy Start	2.78 (1.83, 3.85)	2.36 (1.86, 4.88)
Candidate Local Search Steepest (k=10)	4.31 (3.09, 7.61)	3.97 (3.26, 9.21)
Candidate Local Search Steepest (k=15)	4.21 (3.74, 4.73)	4.17 (3.68, 4.62)
Candidate Local Search Steepest (k=20)	4.92 (4.20, 8.74)	4.47 (3.86, 5.23)
Steepest with LM + Intra Nodes + Random Start	6.11 (4.44, 8.02)	6.43 (4.92, 12.67)
Steepest with LM + Intra Edges + Random Start	4.87 (3.97, 6.03)	4.63 (3.58, 6.56)
Multiple start local search (Steepest, Random init)	2943.55 (2702.39, 3088.22)	2901.23 (2733.71, 2989.75)
Iterated local search	2943.58 (2943.04, 2944.73)	2943.51 (2943.11, 2944.22)
Large Neighborhood Search (without LS)	N\A (set to MSLS mean time)	N\A (set to MSLS mean time)
Large Neighborhood Search (with LS)	N\A (set to MSLS mean time)	N\A (set to MSLS mean time)

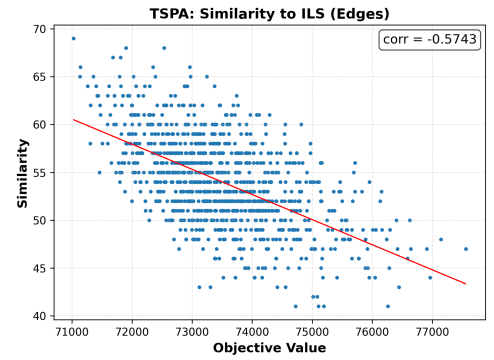
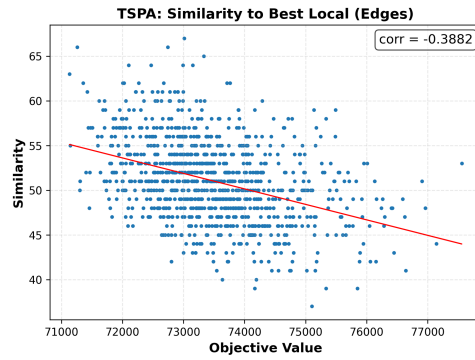
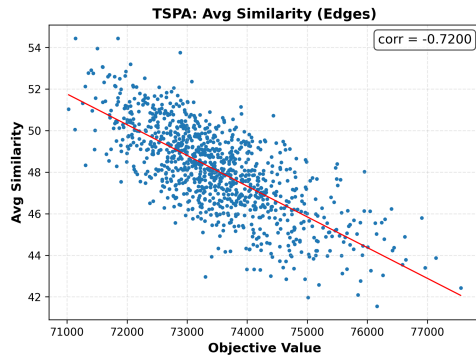
Visualizations

TSPA

Nodes similarity

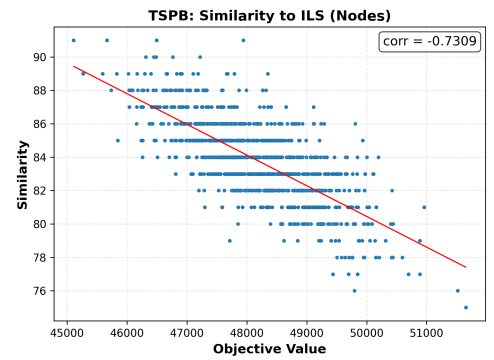
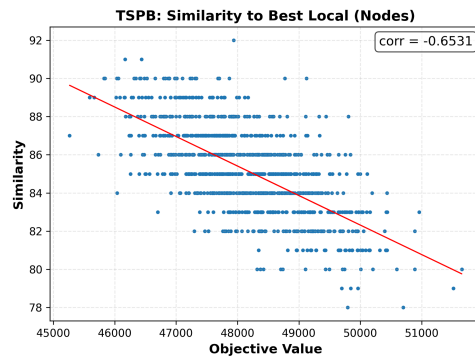
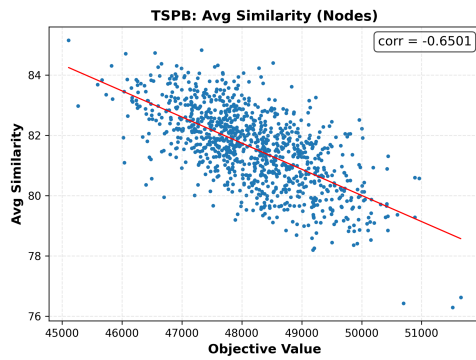


Edges similarity

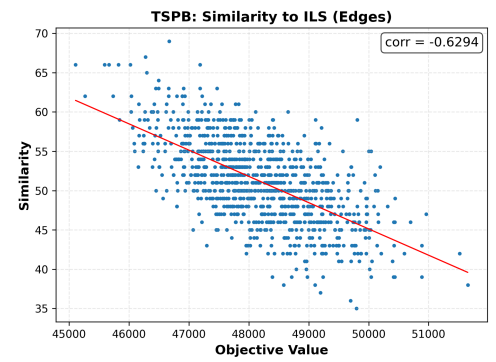
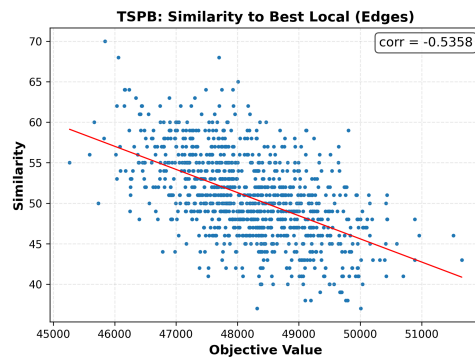
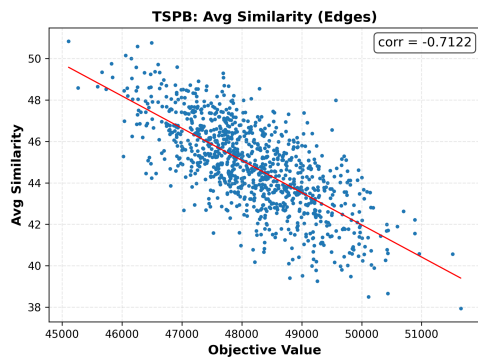


TSPB

Nodes similarity



Edges similarity



Conclusions

- There is a consistent trend where lower-cost solutions have higher similarities. This suggests that solutions closer to optimal are getting more alike each other.